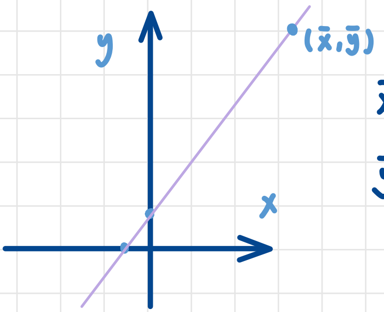
18. $\overset{y}{INSURANCE} = 6.855 + 3.88\ \overset{x}{INCOME}$

se $\quad 7.383 \quad 0.112$

a.



$\bar{x} = 59.3$

$\bar{y} = 6.855 + 3.88 \times 59.3 \approx 236.939$

b. 95% CI : $b_2 \pm t \times se(b_2)$

$b_2 = 3.88$, $df = 20 - 2 = 18$, $t_{0.025,18} \approx 2.101$

$3.88 \pm 2.101 \times 0.112 = [3.645, 4.115]$

c. income = 1000 thousands

$\hat{y} = 6.855 + 3.88 \times 100 = 394.855$

$var(\hat{y}) = var(b_1) + x^2 var(b_2) + 2x\, cov(b_1, b_2)$

$= 7.383^2 + 100^2 \times 0.112^2 + 2 \times 100 \times (-0.746) = 30.7487$

$se(\hat{y}) = \sqrt{30.7487} \approx 5.5452$

$$99\%\ CI : \hat{y} \pm t_{0.005,18}\, se(\hat{y}) = 394.855 \pm \overbrace{2.878 \times 5.5452}^{15.959}$$

$= (378.896\ ,\ 410.814)$

d. $\frac{5000}{1000} = 5$

let $H_0 : \beta_2 = 5$, $H_1 : \beta_2 \neq 5$

$$t = \frac{b_2 - \beta_2}{se(b_2)} = \frac{3.88-5}{0.112} \approx -10$$

reject region : 在 $\alpha = 5\%$ 下, $|t| > 2.101$ 就 reject H_0

$\because |-10| > 2.101$ $\therefore$ reject H_0, there is enough evidences to say $\beta_2 \neq 5$

e. $H_0 : \beta_2 = 1$

$H_1 : \beta_2 > 1$

$$t = \frac{3.88-1}{0.112} \approx 25.71$$

under 1% level of significant

$\because 25.75 > 2.552$ (reject region)

$\therefore$ reject H_0, we have enough evidence to say $\beta_2 > 1$

23. $\overset{y}{PRICE} = \alpha_1 + \alpha_2 \overset{x}{SQFT}^2 + e$

$$\frac{dy}{dx} = 2\alpha_2 x$$

```
Call:
lm(formula = price ~ I(sqft^2), data = collegetown)

Residuals:
    Min      1Q  Median      3Q     Max 
-383.67  -48.39   -7.50   38.75  469.70 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 93.565854   6.072226   15.41   <2e-16 ***
I(sqft^2)    0.184519   0.005256   35.11   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 92.08 on 498 degrees of freedom
Multiple R-squared:  0.7122,	Adjusted R-squared:  0.7117 
F-statistic:  1233 on 1 and 498 DF,  p-value: < 2.2e-16

(a) t-stat: -26.72875 P-value: 1
(b) t-stat: 4.189467 P-value: 1.65395e-05
```

a. $H_0: 40\alpha_1 \le 13$ vs. $H_1: 40\alpha_1 > 13$ $\alpha = 0.05$

$$t = \frac{\hat{a}_2 - 13/40}{se(\hat{a}_2)} = -26.72875 \rightarrow p\text{-val} = 1 > 0.05$$

$$RR = \{ t \ge t_{0.05, 498} \approx 1.648 \}$$

∴ not reject → We don't have sufficient to reject H_0.

We can say the marginal effect on expected house price of increasing the size of a 2000 square feet house by 100 square feet isn't significant greater than \$13000 at 95% CI.

b. $H_0: 80\alpha_1 \le 13$ vs. $H_1: 80\alpha_1 > 13$

$\alpha = 0.05$

$$t = \frac{\hat{a}_2 - 13/80}{se(\hat{a}_2)} = 4.1895 \rightarrow p\text{-val} = 1.654 \times 10^{-5}$$

$$RR = \{ t \ge t_{0.05, 498} \approx 1.648 \}$$

∴ reject → We have sufficient to reject H_0.

We can say the marginal effect on expected house price of increasing the size of a 2000 square feet house by 100 square feet is significant greater than \$13000 at 95% CI.

c. $E(PRICE \mid SQFQ = 20) = 167.3735$

95% CI of $E(PRICE \mid SQFQ = 20) = [158.0481, 176.6988]$

In repeated sampling, about 95% intervals constructed this way will contain the true value of the expected price for a house of 2000 square feet.

d. $\overline{\text{PRICE}}$ under 2000 square feet $= 163.3333$

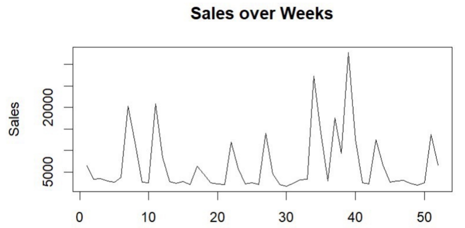
$163.3333 \in [158.0481, 176.6988]$

有落在区間內：代表模型預測与樣本数據相符，模型表現良好

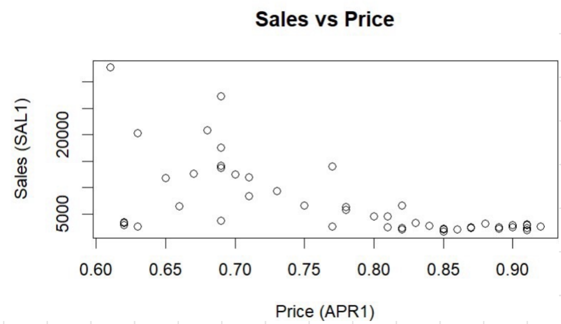
31.

a.

	Mean <dbl>	SD <dbl>	Min <dbl>	Max <dbl>
sal1	6718.7115	6.915083e+03	1762.00	32820.00
apr1	0.7825	9.803711e-02	0.61	0.92



b.



散佈圖應呈負相關，這符合經濟的需求法則，當price上升，購買意願則下降

c.

```
Call:
lm(formula = sal1 ~ price1, data = tuna)

Residuals:
     Min       1Q   Median       3Q      Max
-10869.5  -1588.3   -499.3   1626.2  18607.1

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 40714.22    6196.42   6.571 2.82e-08 ***
price1       -434.45      78.58  -5.528 1.17e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5502 on 50 degrees of freedom
Multiple R-squared:  0.3794,	Adjusted R-squared:  0.367
F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06
```

95% CI for β_2

$[-592.2897, -276.049]$

d. 90% CI for $E[SAL1 \mid PRICE1 = 70] = [8624.934, 11980.87]$

e. $$\text{elasticity} = \frac{\Delta SAL1}{\Delta APR1} \cdot \frac{\overline{APR1}}{\overline{SAL1}} = \frac{1}{100} \cdot \frac{\Delta SAL1}{\Delta PRICE1} \cdot 100\frac{\overline{PRICE1}}{\overline{SAL1}}$$
$$= \beta_2 \frac{\overline{PRICE1}}{\overline{SAL1}}$$

95% CI. of the elasticity: $[-6.8981, -3.2215]$

彈性 >1 → 富有彈性的，因為通常替代品多

f. $H_0: E = -3$

$H_1: E \neq -3$, $\alpha = 0.1$

$$t = \frac{\hat{E} - (-3)}{se(\hat{E})} \overset{H_0}{\sim} t(50) \qquad t = -2.2506$$

$RR = \{ |t| \geq t_{0.05, 50} = 1.6759 \}$

$p\text{-val} = 2 \cdot P(t \leq -2.2506 \mid H_0 \text{ is true}) = 0.0288 \leq 0.1$

→ reject H_0, We have enough evidence to say $E \neq 3$. The elasticity of sales of brand no. 1 with respect to the price of brand no. 1 at the means is significantly different from -3 at 10% significant level.